



CHEMISTRY HSSC-II
SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

Version No.				
4	0	0	5	4

0	●	●	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
●	4	4	4	●
5	5	5	●	5
6	6	6	6	6
7	7	7	7	7
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ROLL NUMBER						

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1	1	1	1	1	1	1
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Deleting/overwriting is not allowed. Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. In a redox reaction, a substance that increases its oxidation number acts as:	Oxidizing agent	Reducing agent	Catalyst	Neutral species	☐	☐	☐	☐
2. If a solute is more soluble in polar solvent than in non-polar solvent, the partition coefficient will:	Be greater than 1	Be less than 1	Be equal to 1	Become zero	☐	☐	☐	☐
3. Which titration curve shows equivalence point above pH 7?	Strong acid vs strong base	Weak acid vs strong base	Strong acid vs weak base	Weak acid vs weak base	☐	☐	☐	☐
4. Which compound is least soluble in water?	MgSO ₄	CaSO ₄	SrSO ₄	BaSO ₄	☐	☐	☐	☐
5. Ligands form bonds with transition metals by:	Accepting electrons	Forming ionic bonds only	Losing protons	Donating lone pairs of electrons	☐	☐	☐	☐
6. Which of the following is a type of stereoisomerism?	Chain isomerism	Position isomerism	Geometrical isomerism	Functional group isomerism	☐	☐	☐	☐
7. Which compound is more stable due to resonance?	Cyclohexene	1,3-Cyclohexadiene	Benzene	Hexane	☐	☐	☐	☐
8. When chloroethane reacts with aqueous OH ⁻ at low temperature, main product is:	Ethene	Ethanol	Benzene	Phenol	☐	☐	☐	☐
9. If the temperature during diazotization rises above 10°C, the diazonium salt:	Becomes more stable	Decomposes rapidly	Converts into alkane	Forms benzene directly	☐	☐	☐	☐
10. In the oxidation reactions side chain of alkyl benzene is converted into:	-COOH group	-OH group	-NH ₂ group	-Cl group	☐	☐	☐	☐
11. Nitriles are prepared from halogenoalkanes by reaction with KCN. This is an example of	Nucleophilic substitution	Elimination reaction	Addition reaction	Polymerization	☐	☐	☐	☐
12. If a polymer segment shows: -CH ₂ -CH ₂ -CH ₂ -CH ₂ -, the monomer is:	Ethene	Propene	Butene	Methane	☐	☐	☐	☐
13. Enzyme activity is affected by:	Density	Colour only	Pressure only	Temperature	☐	☐	☐	☐
14. The molecular ion peak gives information about:	Number of atoms	Colour of compound	Molecular mass of compound	Functional group only	☐	☐	☐	☐
15. Which compound will show maximum absorption in UV/visible region?	Ethene	1,3-Butadiene	Propene	1,3,5-Hexatriene	☐	☐	☐	☐
16. Propanone (CH ₃ -CO-CH ₃) gives how many ¹³ C NMR signals?	4	3	2	1	☐	☐	☐	☐
17. A compound that strongly interacts with the stationary phase will:	Move slower	Move faster	Have R _f = 1	Be colourless	☐	☐	☐	☐

SUPPLEMENTARY TABLE

—2HA-I 26005- (B) —

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40



32

Total Marks Sections B and C: 68

(14x3=42)

SECTION – C (Marks 26)

Q.3	A current of 3.0 A is passed through a solution of CuSO_4 for 45 minutes. (a) Write chemical equation occurs at cathode. (b) Calculate the total charge passed. (c) Calculate the mass of copper deposited. (Molar mass of Cu = 63.5 g mol^{-1} , $z = 2$)	2+2 +3	OR	Explain the SN_1 mechanism of nucleophilic substitution for tert-butyl bromide ($(\text{CH}_3)_3\text{CBr}$) reacting with water. Your answer should: • Show the formation of the carbocation intermediate • Include the role of the inductive effect of alkyl groups in stabilizing the carbocation. Write rate equation for given reaction	3+2 +2
Q.4	What are transition elements? Write trend of melting point and atomic radii from Ti to Cu in 3d series.	1+3 +3	OR	State two factors that affect the basic strength of amines, and use these factors to explain the order of basicity: Phenylamine > Ammonia > Ethylamine.	2+5
Q.5	Explain the mechanism showing the role of the Lewis acid of Friedel-Crafts alkylation of benzene with CH_3Cl using AlCl_3 as a catalyst.	2+4	OR	Explain the trend in thermal stability of Group II carbonates and nitrates, giving reasons. Also, write the order of stability for each.	06
Q.6	Justify the following order of acidic strength by drawing the resonance structures of their conjugate bases: Phenol > Water > Ethanol	3+3	OR	Justify the given order of ease of hydrolysis: Acyl chloride > Alkyl chloride > Aryl chloride.	2+2 +2

- 2HA-I 26005 (B)

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40
Atomic No	21	22	23	24	25	26	27	28	29	30										
3d Series	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn										
Mass No	45	48	51	52	55	56	59	59	64	65										



CHEMISTRY HSSC-II
SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

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Answer Sheet No. _____

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Version No.				
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ROLL NUMBER						

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Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. The maximum distance traveled by the solvent is referred as:		Starting point	Solvent front	R _f value	Mobile phase	☐	☐	☐	☐
2. Catalysts accelerate reactions by:		Increasing temperature only	Changing reaction enthalpy	Increasing reactant concentration	Lowering activation energy	☐	☐	☐	☐
3. $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ What is being oxidized in given reaction?		Cu	Zn^{2+}	Cu^{2+}	Zn	☐	☐	☐	☐
4. Which of the following statements is correct?		Strong bases have high degree of ionization	Weak bases completely ionize	Strong bases have low degree of ionization	Weak bases have high K _b	☐	☐	☐	☐
5. The reactivity of Group 2 elements _____ down the group:		Remains constant	First increases then decreases	Decreases	Increases	☐	☐	☐	☐
6. Zinc is not considered as a transition element because:		d-orbitals are completely filled	It is a non-metal	It has no d-orbitals	It is highly reactive	☐	☐	☐	☐
7. An optically active compound is one that:		Has double bonds	Is colored	Has fruity smell	Rotates plane-polarized light	☐	☐	☐	☐
8. The correct relative reactivity order is:		Alkene > Benzene > Alkane	Alkene > Alkane > Benzene	Benzene > Alkene > Alkane	Alkane > Benzene > Alkene	☐	☐	☐	☐
9. The product formed when chloroethane reacts with KCN is:		Methane	Propanenitrile	Ethanol	Methanenitrile	☐	☐	☐	☐
10. The gas evolved during hydrolysis of diazonium salt is:		Nitrogen	Carbon dioxide	Oxygen	Hydrogen	☐	☐	☐	☐
11. Which of the following is more acidic?		CH_3CH_2COOH	$Cl_2CHCOOH$	$ClCH_2COOH$	CH_3COOH	☐	☐	☐	☐
12. Which amine is more basic in aqueous solution?		$(CH_3)_3N$	$C_6H_5NH_2$	CH_3NH_2	NH_3	☐	☐	☐	☐
13. Which monomers undergo condensation polymerization?		Vinyl chloride	Amino acids	Ethene	Propene	☐	☐	☐	☐
14. Which protein carries oxygen in the blood?		Keratin	Elastin	Hemoglobin	Collagen	☐	☐	☐	☐
15. Identify the $\frac{m}{z}$ value of $CH_3CH_2CH_2^+$:		33	23	43	37	☐	☐	☐	☐
16. The base peak in a mass spectrum corresponds to the:		Molecular ion	Hydrogen bonding	Heaviest fragment	Most stable ion	☐	☐	☐	☐
17. (CH_3CH_2OH) has _____ different carbon environments in its ^{13}C NMR spectrum.		3	2	1	4	☐	☐	☐	☐

—2HA-I 26005- (D)—

SUPPLEMENTARY TABLE

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
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CHEMISTRY HSSC-II

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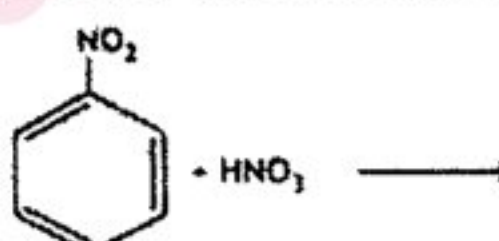
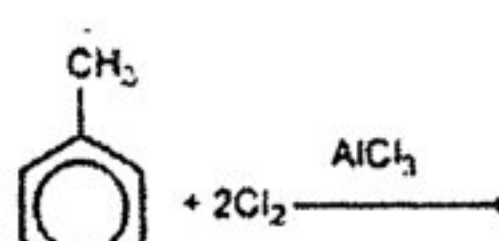
Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly.

(14x3=42)

(i)	Given the following half-cell reactions: $\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$ ($E^\circ = +0.77 \text{ V}$) $\text{Zn}^{2+} + 2e^- \rightarrow \text{Zn}$ ($E^\circ = -0.76 \text{ V}$) identify the oxidation and reduction reactions and calculate E° of cell.	1+1 +1	OR	A solute distributes between water and ether at equilibrium, its concentration in ether is 4 g/L and in water is 2 g/L. Find the partition coefficient (K_{pc}).	03
(ii)	What is Standard Hydrogen Electrode? Write its working with the help of chemical reactions.	1+2	OR	How does bicarbonate buffer system ($\text{H}_2\text{CO}_3/\text{HCO}_3^-$) in blood maintain a constant pH with the help of chemical reactions?	1.5 + 1.5
(iii)	Two bases A and B have K_b values of 1.8×10^{-5} and 1.0×10^{-2} respectively. Identify the stronger base. Justify your answer based on extent of ionization.	1+2	OR	Sketch the titration curve of a weak base with a strong acid and indicate the position of the equivalence point.	2+1
(iv)	Compare the ionization energies of Group 1 and Group 2 elements. Explain the reason for the difference.	1+2	OR	Write the IUPAC name of the following coordination compounds a. $[\text{Pt}(\text{NH}_3)_4\text{BrCl}]\text{SO}_4$ b. $\text{K}_4[\text{Fe}(\text{CN})_6]$	1.5 + 1.5
(v)	Write any three uses of group-II elements in industry, agriculture or medicines.	1x3	OR	Write any three physical properties of transition elements.	1x3
(vi)	Draw the Cis and Trans isomers of 2-butene and identify the stable isomer between them.	2+1	OR	Write the mechanism of nitration of benzene.	03
(vii)	Separate the chiral and Achiral molecules in the following. a. $\text{CF}_3\text{CH}_2\text{CCl}_3$ b. $\text{CF}_2\text{HCHFCCl}_3$ c. $\text{CH}_2\text{ClCH}(\text{OH})\text{CH}_3$	1x3	OR	Identify the major products in the given reactions i  ii 	1.5 + 1.5
(viii)	Draw a pair of enantiomers for 2-butanol and show the chiral center.	2+1	OR	Using retro-synthesis, deduce the starting halogenoalkane for the formation of 2-butanol.	03
(ix)	Compare the stability of primary, secondary, and tertiary carbocations, state the order of stability, and explain the reason for their relative stabilities.	1+1 +1	OR	Arrange water, phenol, and ethanol in order of increasing acidity and explain the reason for this order.	1+2
(x)	Write chemical equations to prepare phenol from aniline.	1.5 + 1.5	OR	Why amides do not readily accept protons like amines? Give a reason	03
(xi)	Differentiate between addition and condensation polymers.	1.5 + 1.5	OR	Briefly explain monosaccharides, disaccharides, and polysaccharides.	1x3
(xii)	Differentiate between molecular ion peak and base peak in mass spectrometry.	1.5 + 1.5	OR	Identify the functional groups (type of bond) corresponding to the following IR absorption bands: $3300\text{--}3500 \text{ cm}^{-1}$, $1700\text{--}1725 \text{ cm}^{-1}$, and $1050\text{--}1150 \text{ cm}^{-1}$.	1x3
(xiii)	How many peaks are expected in the ^1H -NMR spectrum of dimethyl ether? Justify your answer.	1+2	OR	Describe the following terms related to chromatography a. Mobile phase b. R_f value	1.5 + 1.5
(xiv)	Describe any three properties of alloys that make them important.	1x3	OR	Briefly describe any three different types of fertilizers along with their importance.	1x3

SECTION – C (Marks 26)

Note: Attempt the following questions.

Q.3	Balance the given redox equation according to oxidation number method a. $P + HNO_3 + H_2O \rightarrow H_3PO_4 + NO$ b. $HNO_3 + HI \rightarrow NO + I_2 + H_2O$	3+4	OR	What is the trend in thermal stability of Group II carbonates and nitrates? Give reasons. Also, write the order of stability for each.	2+3 +2
Q.4	Compare the arrangement of d-orbitals in octahedral and tetrahedral complexes. (Support your answer by drawing their diagrams of d-d splitting pattern).	3.5 + 3.5	OR	Write the chemical equations for the preparation of Nylon-6,6 and Kevlar. (Use the following monomers: benzene 1,4dioic acid, benzene 1,4-dicarboxylic acid, adipic acid, hexamethylene diamine).	3+4
Q.5	Justify the following order of acidic strength by drawing the resonance structures of their conjugate bases: Ethanoic acid > Phenol > Ethanol	2+2 +2	OR	State two factors that affect the basic strength of amines, and use these factors to explain the order of basicity: Phenylamine > Ammonia > Ethylamine.	2+4
Q.6	When phenol is treated with bromine water, a white precipitate is formed. • Write balanced equation. • Compare this reaction with bromination of benzene	2+4	OR	A student records a 1H NMR spectrum using tetramethylsilane (TMS) as a reference. a. Why TMS is used as a standard for chemical shift measurements? Explain. b. Why deuterated solvents are required in NMR spectroscopy? Explain.	3+3

— 2HA-I 26005 (D) —

SUPPLEMENTARY TABLE

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(Chemistry Page 2 of 2)